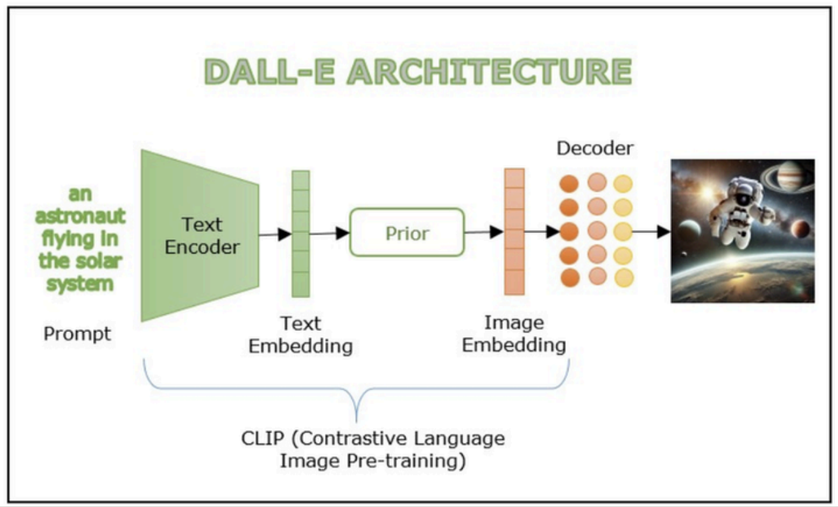
MULTIMODEL:- IMAGE TO TEXT

- A MULTIMODAL MODEL IS AN AI THAT CAN UNDERSTAND AND WORK WITH MORE THAN ONE TYPE OF DATA LIKE IMAGES AND TEXT AT THE SAME TIME.

- OBJECTIVE:- CAN WE TEACH A FROZEN LANGUAGE MODEL (GPT-2) TO DESCRIBE IMAGES WITHOUT RETRAINING IT FROM SCRATCH, BY SIMPLY BRIDGING IT TO A VISUAL ENCODER (CLIP) USING LIGHTWEIGHT ADAPTER LAYERS?



- CLIP IS A VISUAL MODEL TRAINED BY OPENAI THAT CONVERTS ANY IMAGE INTO A VECTOR REPRESENTING WHAT IT SEES.
- GPT-2 IS A LANGUAGE MODEL TRAINED ON BILLIONS OF WEB SENTENCES THAT GENERATES FLUENT TEXT ONE WORD AT A TIME, PREDICTING THE MOST LIKELY NEXT WORD.

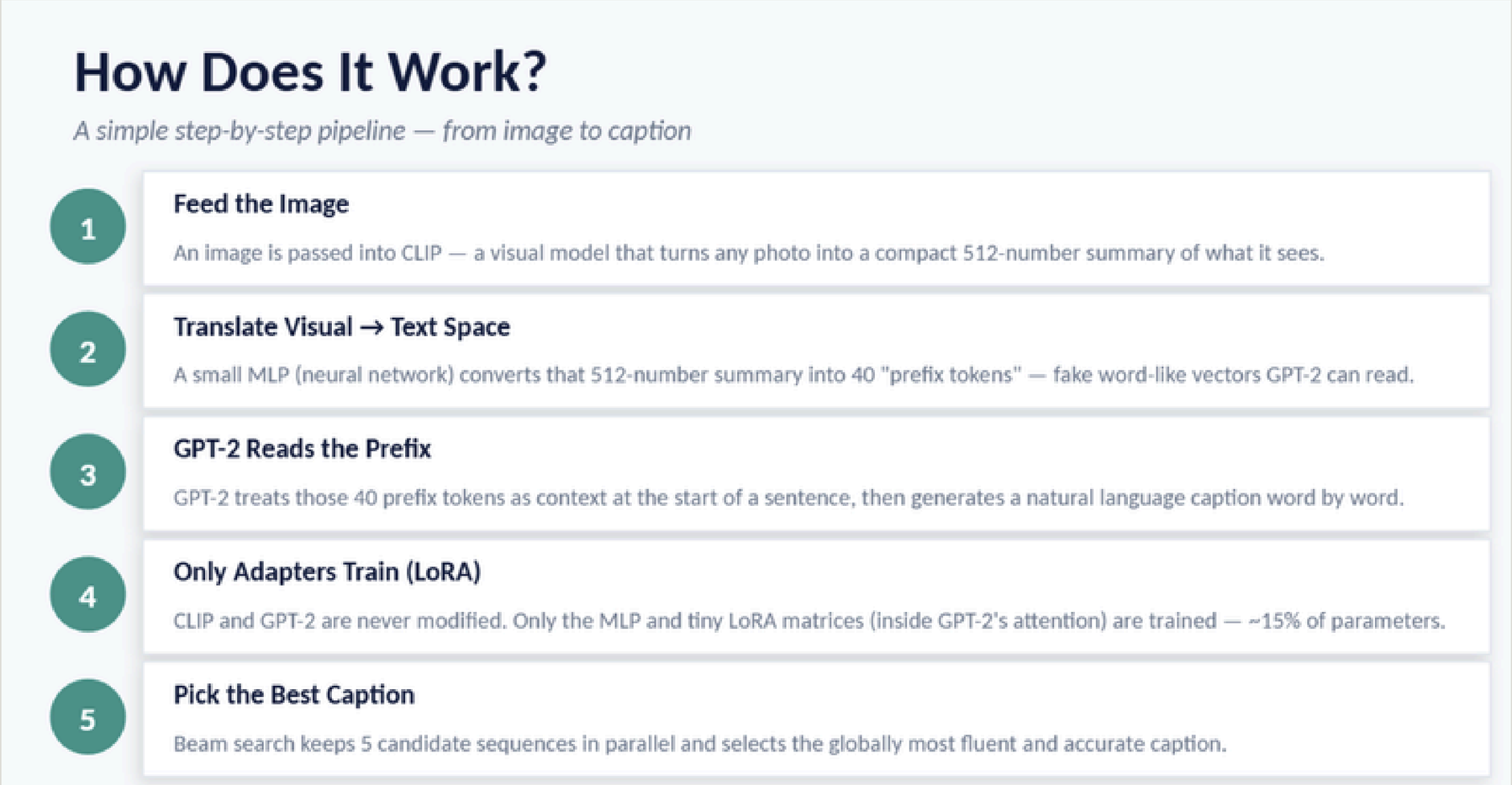
- CLIP PRODUCE A 512-DIMENSIONAL VECTOR (THINK OF IT LIKE A FINGUREPRINT WHERE IT STORES ALL THE INFORMATION ABOUT THE IMAGE IN THOSE 512 VECTOR POINTS)

- GPT-2 DOESN'T ACTUALLY READ 512 NUMBERS DIRECTLY — IT WORKS IN 768-DIMENSIONAL TOKEN SPACE (EACH "WORD" IT READS IS A LIST OF 768 NUMBERS)

What are we training?

- the MLP Projection layer is a small 2 layer neural network we train from scratch that takes the 512-d CLIP vector and convertrs it to 40 x 768-d prefix tokens
- These 40 tokens are called prefix tokens — fake word-like vectors that GPT-2 reads as if they were the beginning of a sentence

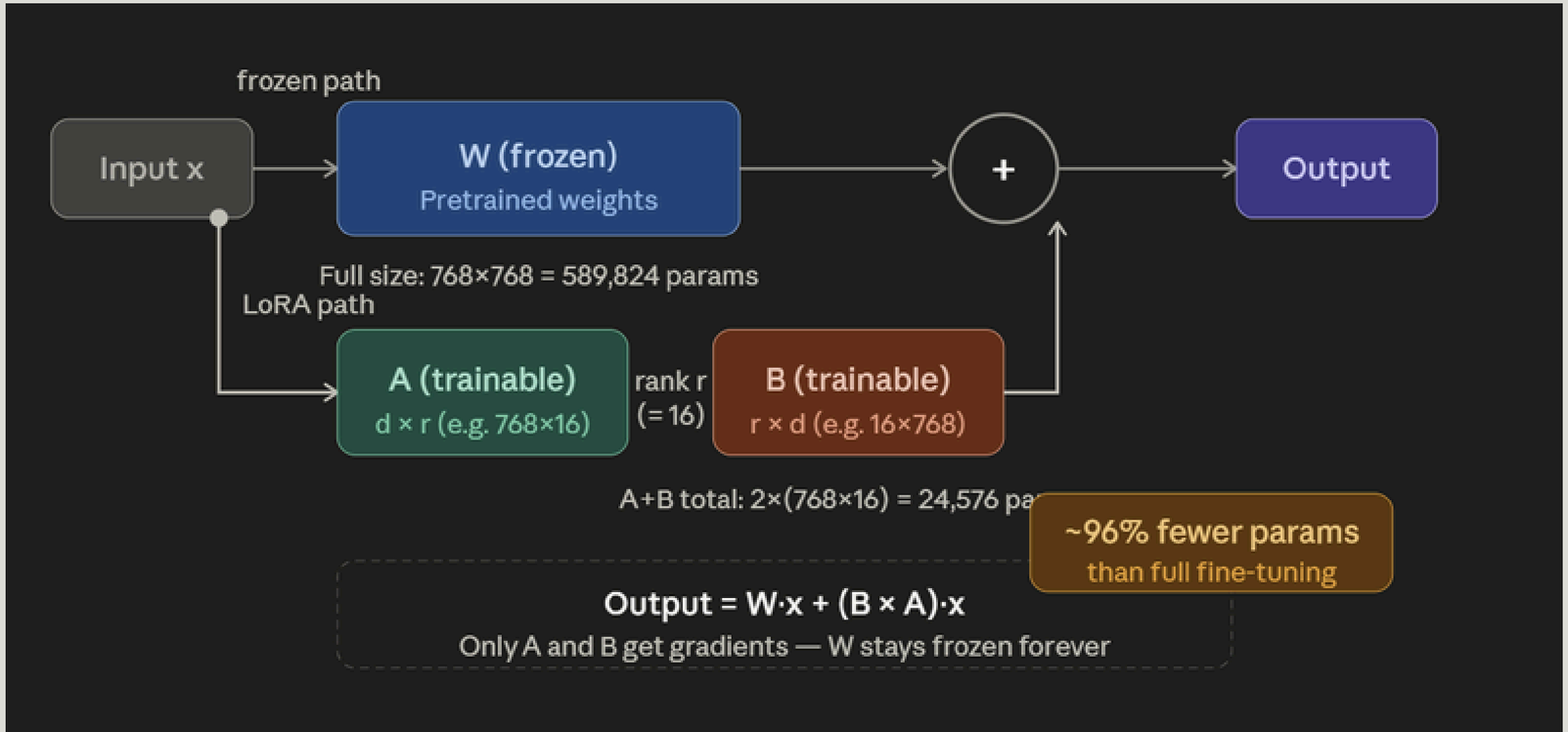
THEREFORE IN SHORT CLIP speaks in 512 numbers, GPT-2 speaks in 768 numbers per word, and the MLP is the trained translator between the two



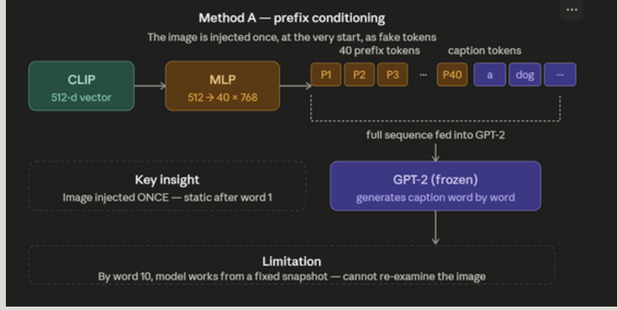
LORA (LOW-RANK ADAPTATION)

LORA (LOW-RANK ADAPTATION) IS A SMART WAY TO FINE-TUNE A LARGE MODEL WITHOUT CHANGING ALL OF ITS WEIGHTS. THE PROBLEM IT SOLVES IS THAT GPT-2 HAS MILLIONS OF PARAMETERS AND UPDATING ALL OF THEM IS SLOW AND EXPENSIVE

HOW LORA WORKS IS THAT INSTEAD OF MODIFYING GPT-2'S ORIGINAL WEIGHT MATRICES, LORA FREEZES THEM AND INJECTS TWO TINY MATRICES A AND B ALONGSIDE THEM AND IN OUR PROJECT ONLY 15% OF THE TOTAL PARAMETERS ARE TRAINED AND DOING SO THE MODEL LEARNED TO CAPTION IMAGES EFFICIENTLY WITHOUT LOSING ITS LANGUAGE FLUENCY



HOW DO WE INJECT THE IMAGE INTO GPT-2



METHOD A — PREFIX CONDITIONING

- IMAGE IS COMPRESSED INTO ONE 512-D VECTOR → MLP PROJECTS IT INTO 40 PREFIX TOKENS → PREPENDED TO THE SENTENCE
- GPT-2 READS THESE 40 TOKENS AS A PREAMBLE, THEN GENERATES THE CAPTION — THE IMAGE IS INJECTED ONCE AT THE START
- SIMPLE, DATA-EFFICIENT, AND WORKS WELL WITH LIMITED TRAINING DATA (25K–100K PAIRS)
- LIMITATION: THE VISUAL SIGNAL IS STATIC — BY WORD 10 OF THE CAPTION, THE MODEL CAN'T "RE-LOOK" AT THE IMAGE

METHOD B — CROSS-ATTENTION CONDITIONING

- INSTEAD OF INJECTING ONCE, A CROSS-ATTENTION LAYER IS ADDED AFTER EVERY GPT-2 SELF-ATTENTION BLOCK (12 LAYERS TOTAL)
- CLIP'S 50 RAW PATCH VECTORS (SPATIAL REGIONS OF THE IMAGE) ARE PASSED AS KEYS & VALUES
- AT EVERY WORD GENERATION STEP, THE MODEL ACTIVELY 'LOOKS' AT THE RELEVANT IMAGE PATCHES — LIKE QUERYING THE IMAGE FOR EACH WORD
- MORE POWERFUL IN THEORY, BUT REQUIRES FAR MORE TRAINING DATA TO LEARN THE PATCH-TO-WORD ALIGNMENT

Decoding Strategies

How does the model decide which word to generate next?

⚡ Greedy Search	Always pick the single highest-probability next token — fast, but often produces repetitive or generic captions.
Beam Search (k=5) Used	Keep the top 5 candidate sequences in parallel; pick the globally best complete caption at the end — balances quality and compute.
Nucleus (Top-p) Sampling	Randomly sample from the smallest set of tokens whose cumulative probability exceeds p — adds diversity at the cost of occasional randomness.
🌡 Temperature Scaling	Divide logits by a temperature value before softmax — lower values make output more focused and conservative, higher values increase diversity.
Contrastive Decoding (α=0.5)	Subtract the unconditional language score from the visual score at each step — penalises hallucinations caused by language priors.

THE REASON WE CHOSE BEAM SEARCH OVER THE OTHER STRATEGIES IS CONSISTENCY AND REPRODUCIBILITY

OUR MODEL

v1 · Baseline	Linear projection | Prefix length 10 | LoRA rank 8 | 5k images — simple starting point, just a straight-line mapping from CLIP to GPT-2.
v2 · Non-linear	MLP + Tanh projection | Prefix 20 | LoRA rank 8 — added a non-linear projection so the bridge can learn more complex visual-to-language mappings.
v3 · Full Capacity	MLP + Tanh | Prefix 40 | LoRA rank 16 | 5k COCO val images on T4 GPU — doubled capacity to give GPT-2 a richer visual context.
v4 · Primary Result	Same arch as v3 | Scaled to 20k train images (100k pairs) | Batch 64 on H100 — CIDEr 0.7668, BLEU-4 0.2582, beats BLIP by ~4.9%.
Method B / B+	Cross-attention injection using 50 CLIP patch vectors per layer | B+: ViT-L/14 + Gated Attn + LoRA r32 — powerful architecture, underperformed due to data constraints.

TABLE I
QUANTITATIVE COMPARISON OF ALL EXPERIMENTAL VER

Version	Pairs	B-1	B-4	MTR	R-L	CIDEr
v3	25k	0.683	0.254	0.253	0.497	0.751
v4	**100k**	**0.695**	**0.258**	**0.262**	**0.497**	**0.767**
Method B	100k	0.576	0.194	0.399	0.395	0.552
Method B+	25k	0.633	0.225	0.253	0.431	0.605
BLIP	129M	0.599	0.250	0.229	0.426	0.731

B-1/B-4=BLEU; MTR=METEOR; R-L=ROUGE-L.
Method B+: ViT-L/14 + Gated Cross-Attn + LoRA r32.

WHY METHOD A BEAT METHOD B:

- METHOD A (PREFIX) WORKS WELL WITH LIMITED DATA BECAUSE IT LEANS HEAVILY ON GPT-2'S EXISTING PRETRAINED LANGUAGE KNOWLEDGE
- THE MLP JUST NEEDS TO LEARN ONE SIMPLE TRANSLATION FROM VISUAL TO LANGUAGE SPACE.
- METHOD B (CROSS-ATTENTION) INTRODUCES 12 BRAND NEW RANDOMLY INITIALIZED ATTENTION LAYERS THAT HAVE TO LEARN FROM SCRATCH WHICH IMAGE PATCH IS RELEVANT FOR EVERY SINGLE WORD GENERATED (A MUCH HARDER PROBLEM THAT NEEDS FAR MORE DATA TO SOLVE).
- WITH ONLY 100K TRAINING PAIRS, THOSE CROSS-ATTENTION LAYERS NEVER CONVERGED PROPERLY, SO METHOD A WON DESPITE BEING THE SIMPLER ARCHITECTURE.

LLaVA (Large Language and Vision Assistant) is a modern vision-language model that combines a CLIP visual encoder with a large 7 billion parameter language model (Mistral 7B), trained end-to-end on hundreds of millions of image-text pairs.

THANK YOU

GROUP 10

ROHAN KOLLA

SADHVIKA REDDY ANDEM

VENKATA KASHYAP AKULA

ARVIND